

CLASS - XI
MATHEMATICS FORMULA LIST

Geometry

- In a clock, a minute hand rotates through an angle of 6° in one minute.
- In a clock, an hour hand rotates through an angle of $\left(\frac{1}{2}\right)^\circ$ in one minute.

Set Theory

The total number of subsets of a finite set consisting of n elements is 2^n .

For any three sets A, B and C :

- | | |
|--|---|
| (i) $(A \cup B) \cup C = A \cup (B \cup C)$ | (ii) $(A \cap B) \cap C = A \cap (B \cap C)$ |
| (iii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ | (iv) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ |
| (v) $(A \cup B)' = A' \cap B'$ | (vi) $(A \cap B)' = A' \cup B'$ (De Morgan's Law) |
| (vii) $A - (B \cup C) = (A - B) \cap (A - C)$ | (viii) $A - (B \cap C) = (A - B) \cup (A - C)$ |

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A - B) = n(A) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Number of elements in exactly two of the sets A, B and C =

$$n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$$

Number of elements in exactly one of the sets A, B and C =

$$n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(B \cap C) - 2n(C \cap A) + 3n(A \cap B \cap C)$$

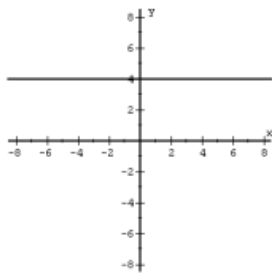
Relations and Functions

For any three sets A, B and C :

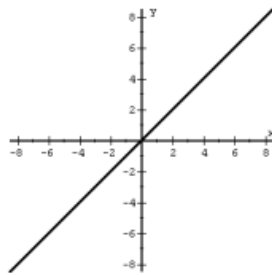
- | | |
|--|---|
| (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$ | (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$ |
| (iii) $(A - B) \times C = (A \times C) - (B \times C)$ | (iv) $(A \cap B) \times C = (A \times C) \cap (B \times C)$ |

If A and B are two non-empty finite sets consisting of m and n elements respectively, then the total number of relations from A to B is 2^{mn}

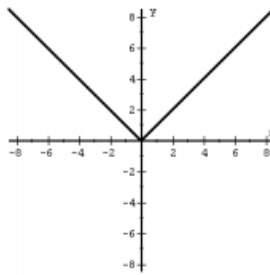
PARENT FUNCTIONS



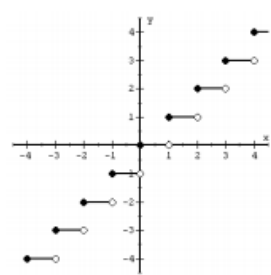
$f(x) = a$
Constant



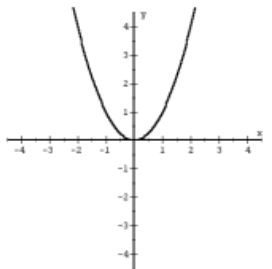
$f(x) = x$
Linear



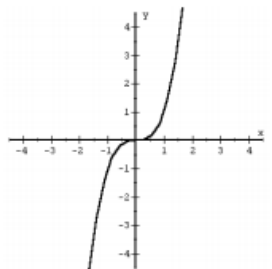
$f(x) = |x|$
Absolute Value



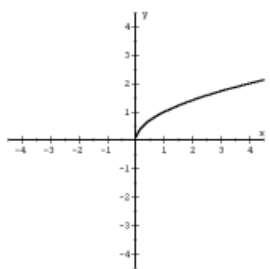
$f(x) = \text{int}(x) = [x]$
Greatest Integer



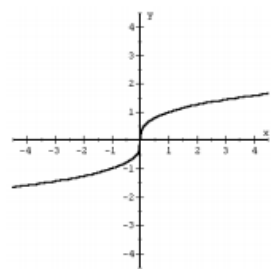
$f(x) = x^2$
Quadratic



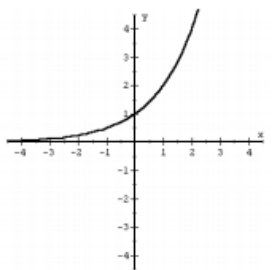
$f(x) = x^3$
Cubic



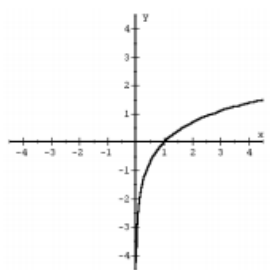
$f(x) = \sqrt{x}$
Square Root



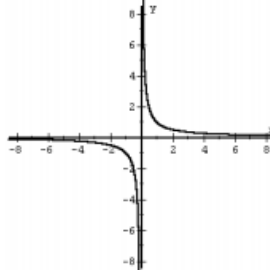
$f(x) = \sqrt[3]{x}$
Cube Root



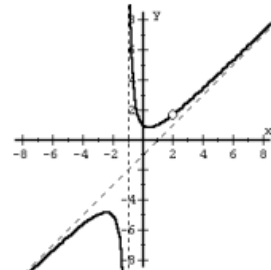
$f(x) = a^x$
Exponential



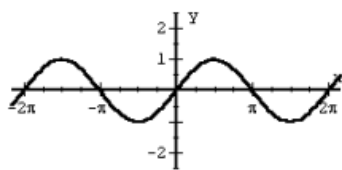
$f(x) = \log_a x$
Logarithmic



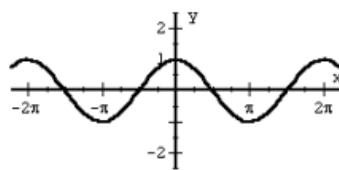
$f(x) = \frac{1}{x}$
Reciprocal



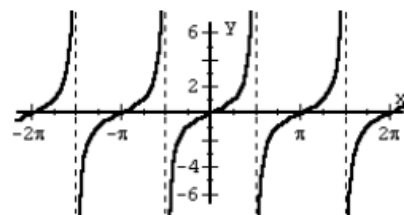
$f(x) = \frac{(x^2 + 1)(x - 2)}{(x + 1)(x - 2)}$
Rational



$f(x) = \sin x$



$f(x) = \cos x$



$f(x) = \tan x$

Trigonometric Functions

Trigonometry

$$180^\circ = \pi \text{ radians} \quad ; \quad \theta = \frac{l}{r}, \theta \text{ is measured in radians}$$

$$\text{In a right triangle:} \quad \sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

Pythagoras' Theorem :

In a right triangle ABC , where a, b are the short sides and c is the hypotenuse then $c^2 = a^2 + b^2$

Trigonometric Ratios of Special Angles

$$\begin{array}{lll} \cos 45^\circ = \frac{1}{\sqrt{2}} & \cos 60^\circ = \frac{1}{2} & \cos 30^\circ = \frac{\sqrt{3}}{2} \\ \sin 45^\circ = \frac{1}{\sqrt{2}} & \sin 60^\circ = \frac{\sqrt{3}}{2} & \sin 30^\circ = \frac{1}{2} \\ \tan 45^\circ = 1 & \tan 60^\circ = \sqrt{3} & \tan 30^\circ = \frac{1}{\sqrt{3}} \end{array}$$

Trigonometric Ratios of Complementary Angles

$$\begin{array}{ll} \sin(90^\circ - \theta) = \cos \theta & ; \quad \cos(90^\circ - \theta) = \sin \theta \\ \tan(90^\circ - \theta) = \cot \theta & ; \quad \cot(90^\circ - \theta) = \tan \theta \\ \sec(90^\circ - \theta) = \operatorname{cosec} \theta & ; \quad \operatorname{cosec}(90^\circ - \theta) = \sec \theta \end{array}$$

Signs of Trigonometric Ratios in Quadrants

2nd quadrant	1st quadrant
S (sin)	A (all)
T (tan)	C (cos)
3rd quadrant	4th quadrant

$$\cos(-\theta) = \cos \theta \quad ; \quad \sin(-\theta) = -\sin \theta \quad ; \quad \tan(-\theta) = -\tan \theta$$

$$\sec \theta = \frac{1}{\cos \theta} \quad ; \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad ; \quad \cot \theta = \frac{1}{\tan \theta} \quad ; \quad \tan \theta = \frac{\sin \theta}{\cos \theta} \quad ; \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1 \quad ; \quad 1 + \tan^2 \theta = \sec^2 \theta \quad ; \quad 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

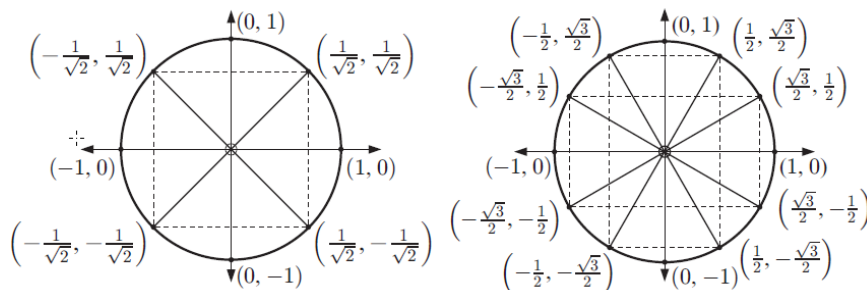
$$-1 \leq \sin x \leq 1 \quad ; \quad -1 \leq \cos x \leq 1 \quad ; \quad -\infty < \tan x < \infty$$

Degrees	0	45	90	135	180	225	270	315	360
Radians	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π

Degrees	0	45	90	135	180	225	270	315	360
Radians	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π

θ (degrees)	0°	90°	180°	270°	360°	450°
θ (radians)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π	$\frac{5\pi}{2}$
sine	0	1	0	-1	0	1
cosine	1	0	-1	0	1	0
tangent	0	undef	0	undef	0	undef

θ (degrees)	30°	45°	60°	135°	150°	240°	315°
θ (radians)	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	$\frac{4\pi}{3}$	$\frac{7\pi}{4}$
sine	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{2}}$
cosine	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
tangent	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	$\sqrt{3}$	-1



$$\sin(A+B) = \sin A \cos B + \cos A \sin B \quad ; \quad \sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B \quad ; \quad \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \quad ; \quad \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$$

$$\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

$$2\sin A \cos B = \sin(A+B) + \sin(A-B) \quad ; \quad 2\cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2\cos A \cos B = \cos(A+B) + \cos(A-B) \quad ; \quad 2\sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) \quad ; \quad \sin A - \sin B = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) \quad ; \quad \cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$\sin 2A = 2\sin A \cos A = \frac{2\tan A}{1 + \tan^2 A}$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\sin A = \pm \sqrt{\frac{1 - \cos 2A}{2}} \quad ; \quad \cos A = \pm \sqrt{\frac{1 + \cos 2A}{2}}$$

$$\sin 3A = 3\sin A - 4\sin^3 A \quad ; \quad \cos 3A = 4\cos^3 A - 3\cos A \quad ; \quad \tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$$

Sine Rule

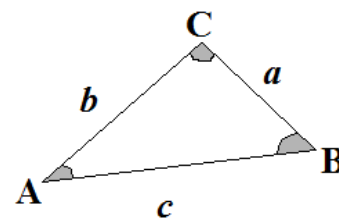
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k$$

Cosine Rule

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad ; \quad \cos B = \frac{a^2 + c^2 - b^2}{2ac} \quad ; \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Area of triangle formula:

$$= \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A = \frac{1}{2} ac \sin B$$



Maximum and minimum values of $a \cos \theta + b \sin \theta$ are $\sqrt{a^2 + b^2}$ and $-\sqrt{a^2 + b^2}$ respectively.

Trigonometric Equation

(i) $\sin \theta = 0$

(ii) $\cos \theta = 0$

(iii) $\tan \theta = 0$

(iv) $\sin \theta = \sin \alpha$

(v) $\cos \theta = \cos \alpha$

(vi) $\tan \theta = \tan \alpha$

(vii)
$$\left. \begin{aligned} \sin^2 \theta &= \sin^2 \alpha \\ \cos^2 \theta &= \cos^2 \alpha \\ \tan^2 \theta &= \tan^2 \alpha \end{aligned} \right\}$$

General Solution

$$\theta = n\pi, n \in \mathbb{Z}$$

$$\theta = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

$$\theta = n\pi, n \in \mathbb{Z}$$

$$\theta = n\pi + (-1)^n \alpha, n \in \mathbb{Z}$$

$$\theta = 2n\pi \pm \alpha, n \in \mathbb{Z}$$

$$\theta = n\pi + \alpha, n \in \mathbb{Z}$$

$$\theta = n\pi \pm \alpha, n \in \mathbb{Z}$$

$$\sin n\pi = 0 \quad ; \quad \cos n\pi = (-1)^n \quad ; \quad \tan n\pi = 0$$

Complex Numbers

Multiplicative Inverse of $z = a + ib$ is $\frac{\bar{z}}{|z|^2}$.

Conjugate of $z = a + ib$ is $\bar{z} = a - ib$.

Modulus of $z = a + ib$ is $|z| = \sqrt{a^2 + b^2}$

For three complex numbers z_1, z_2, z_3 :

(i) $(\bar{\bar{z}}) = z$ (ii) $z + \bar{z} = 2\operatorname{Re}(z)$ (iii) $z - \bar{z} = 2i\operatorname{Im}(z)$ (iv) $z \bar{z} = |z|^2$

(v) $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$ (vi) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2} \quad ; \quad z_2 \neq 0$

Square root of a complex number:

$$\text{Use } (a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$$

$$\sqrt{z} = \pm \left\{ \sqrt{\frac{|z| + \operatorname{Re}(z)}{2}} + i \sqrt{\frac{|z| - \operatorname{Re}(z)}{2}} \right\} ; \text{ if } \operatorname{Im}(z) > 0$$

$$\sqrt{z} = \pm \left\{ \sqrt{\frac{|z| + \operatorname{Re}(z)}{2}} - i \sqrt{\frac{|z| - \operatorname{Re}(z)}{2}} \right\} ; \text{ if } \operatorname{Im}(z) < 0$$

$$\text{Polar Form: } z = r(\cos \theta + i \sin \theta)$$

Algorithm:

Step 1: Find $\alpha = \tan^{-1} \left| \frac{y}{x} \right|$, lying between 0 and $\frac{\pi}{2}$.

Step 2: Determine in which quadrant the point $P(x, y)$ belongs.

If **I quad** then $\theta = \alpha$, If **II quad** then $\theta = \pi - \alpha$

If **III quad** then $\theta = \pi + \alpha$, If **IV quad** then $\theta = 2\pi - \alpha$

Quadratic Formula

$$\text{If } ax^2 + bx + c = 0, \text{ then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Linear In equations

$$(i) x^2 \leq a^2 \Leftrightarrow |x| \leq a \Leftrightarrow -a \leq x \leq a$$

$$(ii) x^2 \geq a^2 \Leftrightarrow |x| \geq a \Leftrightarrow x \leq -a \text{ or } x \geq a$$

$$(iii) x^2 < a^2 \Leftrightarrow |x| < a \Leftrightarrow -a < x < a$$

$$(iv) x^2 > a^2 \Leftrightarrow |x| > a \Leftrightarrow x > a \text{ or } x < -a$$

$$(v) |x - a| < b \Leftrightarrow a - b < x < a + b$$

$$(vi) |x - a| > b \Leftrightarrow x > a + b \text{ or } x < a - b$$

Permutation and Combination

$${}^nP_r = P(n, r) = \frac{n!}{(n-r)!} ; \quad {}^nC_r = C(n, r) = \frac{n!}{r!(n-r)!} ; 1 \leq r \leq n$$

Fundamental Principle of multiplication: If there are two jobs that one of them can be completed in m ways, and when it has been completed, second job can be completed in n ways; then the two jobs in succession can be completed in $m \times n$ ways.

$$\text{Number of all permutations of } n \text{ distinct things taken } r \text{ at a time} = {}^nP_r = P(n, r) = \frac{n!}{(n-r)!}$$

$$\text{Number of all permutations of } n \text{ distinct things taken all at a time} = n!$$

Number of mutually distinguishable permutations of n things, taken all at a time, of which p

are alike of one kind, q are of second kind such that $p + q = n$, is $\frac{n!}{p!q!}$.

Number of all combinations of n distinct things taken r at a time $= {}^nC_r = C(n, r) = \frac{n!}{r!(n-r)!}$.

$${}^nC_0 = {}^nC_n = 1 \quad ; \quad {}^nC_r = {}^nC_{n-r} \quad ; \quad \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

If ${}^nC_x = {}^nC_y$, then $x = y$ or $x + y = n$

If n is an even natural number, then greatest of the values ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ is ${}^nC_{n/2}$.

If n is an odd natural number, then the greatest of the values ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ is ${}^nC_{n-1/2}$.

Binomial Theorem

$$(x+a)^n = {}^nC_0 x^n a^0 + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + {}^nC_3 x^{n-3} a^3 + \dots + {}^nC_n a^n = \sum_{r=0}^n {}^nC_r x^{n-r} a^r$$

Finding specific term of $(a+b)^n$, $T_{r+1} = {}^nC_r a^{n-r} b^r$

$$(1+x)^n = {}^nC_0 x^0 + {}^nC_1 x^1 + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n$$

Finding specific term of $(1+x)^n$, $T_{r+1} = {}^nC_r x^r$.

$$(x-a)^n = {}^nC_0 x^n a^0 - {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 - {}^nC_3 x^{n-3} a^3 + \dots + (-1)^n {}^nC_n a^n = \sum_{r=0}^n (-1)^r {}^nC_r x^{n-r} a^r$$

$$(x+a)^n + (x-a)^n = 2 \left[{}^nC_0 x^n a^0 + {}^nC_2 x^{n-2} a^2 + {}^nC_4 x^{n-4} a^4 \dots \right]$$

$$(x+a)^n - (x-a)^n = 2 \left[{}^nC_1 x^{n-1} a + {}^nC_3 x^{n-3} a^3 + {}^nC_5 x^{n-5} a^5 \dots \right]$$

If n is odd then, $(x+a)^n + (x-a)^n$ and $(x+a)^n - (x-a)^n$ will have $\left(\frac{n+1}{2}\right)$ terms.

If n is even then, $(x+a)^n + (x-a)^n$ have $\left(\frac{n}{2}+1\right)$ terms and $(x+a)^n - (x-a)^n$ will have $\left(\frac{n}{2}\right)$ terms.

In expansion of $(x+a)^n$, the r^{th} term from the end is $(n-r+2)^{\text{th}}$ term from beginning.

Middle term in Binomial Expansion:

Since $(x+a)^n$ contains $(n+1)$ terms, therefore,

(i) If n is even, then $\left(\frac{n}{2}+1\right)$ th term is the middle term.

(ii) If n is odd, then $\left(\frac{n+1}{2}\right)$ th and $\left(\frac{n+3}{2}\right)$ th terms are the two middle terms.

Arithmetic Progressions

The **n th term of an A.P.** with first term ' a ' and common difference ' d ' is given by

$$a_n = a + (n-1)d$$

The **sum to n terms of an A.P.** with first term ' a ' and common difference ' d ' is given by

$$S_n = \frac{n}{2} [2a + (n-1)d] = \frac{n}{2} (a+l) \quad \text{where } l \text{ is last term}$$

Insertion of n Arithmetic Means between a and b :

Let $A_1, A_2, A_3, \dots, A_n$ are inserted. Then,

$$A_1 = a+d, A_2 = a+2d, A_3 = a+3d, \dots, A_n = a+nd ; \text{ where } d = \frac{b-a}{n+1}$$

Ways of selecting terms of an AP:

Number of terms	Terms
3	$a-d, a, a+d$
4	$a-3d, a-d, a+d, a+3d$

Geometric Progressions

The **n th term of an G.P.** with first term ' a ' and common ratio r is given by $a_n = ar^{n-1}$

The **sum to n terms of an G.P.** with first term ' a ' and common ratio r is given by

$$S_n = a \left(\frac{r^n - 1}{r - 1} \right) = a \left(\frac{1 - r^n}{1 - r} \right) ; r \neq 1$$

Ways of selecting terms of an GP :

Number of terms	Terms
3	$\frac{a}{r}, a, ar$
4	$\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$

Insertion of n Geometric Means between a and b :

Let $G_1, G_2, G_3, \dots, G_n$ are inserted. Then,

$$G_1 = ar, G_2 = ar^2, G_3 = ar^3, \dots, G_n = ar^n ; \text{ where } r = \left(\frac{b}{a} \right)^{\frac{1}{n+1}}$$

If A and G be the A.M. and G.M. between two positive numbers, then the numbers are

$$A \pm \sqrt{A^2 - G^2}.$$

Sum of an infinite GP with first term ' a ' and common ratio r is given by $S_\infty = \frac{a}{1-r} ; -1 < r < 1$

Sum to n terms of some special series:

$$(i) 1+2+3+\dots+n = \frac{n(n+1)}{2}$$

$$(ii) 1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$(iii) 1^3+2^3+3^3+\dots+n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

$$*(iv) 1^4+2^4+3^4+\dots+n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

$$S_n = \sum_{r=1}^n T_r$$

Straight Lines

Slope of line, m , through the points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$

Forms of Linear Equations

Point-slope form

$$y - y_1 = m(x - x_1)$$

Slope-intercept form

$$y = mx + b$$

General form

$$Ax + By + C = 0$$

Horizontal line

$$y = c$$

Vertical line

$$x = c$$

Distance, d , between the points (x_1, y_1) and (x_2, y_2) is given by $d = \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$

Midpoint Formula

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Section Formula

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

The area of the triangle formed by the points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ is given by

$$\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Coordinates of the **Centroid** of the triangle whose vertices are

$$A(x_1, y_1), B(x_2, y_2) \text{ and } C(x_3, y_3) \text{ are } \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

*Coordinates of the **In-Centre** of the triangle whose vertices are

$$A(x_1, y_1), B(x_2, y_2) \text{ and } C(x_3, y_3) \text{ are } \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right) \text{ where}$$

$$a = BC, b = CA, c = AB.$$

Slope of line $ax + by + c = 0$ is $-\frac{\text{coeff. of } x}{\text{coeff. of } y} = -\frac{a}{b}$

An acute angle θ between two lines having slopes m_1 and $m_2 =$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| ; 1 + m_1 m_2 \neq 0$$

Lines are parallel if $m_1 = m_2$ and Lines are perpendicular if $m_1 m_2 = -1$

Equation of line which passes through the point (x_1, y_1) and has a slope m is $y - y_1 = m(x - x_1)$.

Equation of line which passes through the points (x_1, y_1) and (x_2, y_2) is $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Equation of line making intercepts a and b on x and y axis respectively is $\frac{x}{a} + \frac{y}{b} = 1$.

Equation of line upon which the length of perpendicular from the origin is p and the angle between this perpendicular and positive x -axis is α is given by $x \cos \alpha + y \sin \alpha = p$.

Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

The perpendicular distance (d) of a line $ax + by + c = 0$ from a point $(x_1, y_1) = \frac{|a_1x + b_1y + c|}{\sqrt{a^2 + b^2}}$.

The equation of line passing through (x_1, y_1) and making angle θ with the positive direction

of x -axis is $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$, where r is the distance of the point (x, y) on the line from the point (x_1, y_1) .

Conic Section

Let l be a fixed line and F be a fixed point not on l , and $e > 0$ be a fixed real number. Let $|MP|$ be the perpendicular distance from a point P (in the plane of the line l and point F) to the line l , then the locus of all points P such that $|FP| = e |MP|$ is called a *conic*.



The fixed point F is called a **focus** of the conic and the fixed line l is called the **directrix** associated with F . The fixed real number e (> 0) is called eccentricity of the conic. In particular, a conic with eccentricity e is called

- (i) a **parabola** iff $e = 1$
- (ii) an **ellipse** iff $e < 1$
- (iii) a **hyperbola** iff $e > 1$.

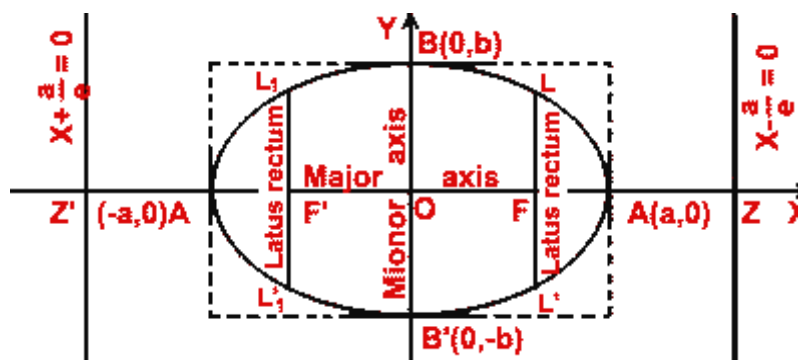
Main facts about the Parabola

Equations	$y^2 = 4ax$, ($a > 0$) Right hand	$y^2 = -4ax$, $a > 0$ Left hand	$x^2 = 4ay$, $a > 0$ Upwards	$x^2 = -4ay$, $a > 0$ Downwards
Axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Eqn. of Directrix	$x + a = 0$	$x - a = 0$	$y + a = 0$	$y - a = 0$
Focus	$(a, 0)$	$(-a, 0)$	$(0, a)$	$(0, -a)$
Vertex	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
Length of Latus-rectum	$4a$	$4a$	$4a$	$4a$

Main facts about the Ellipse

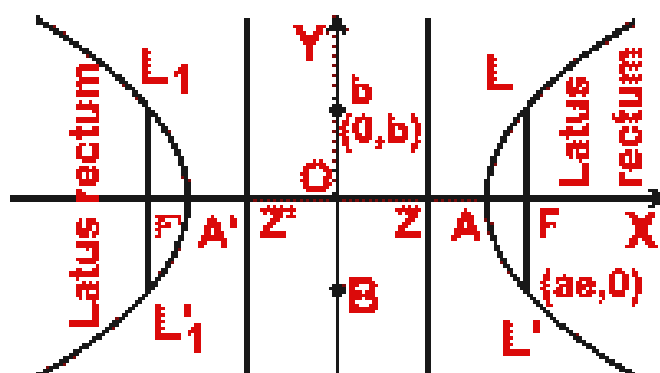
Equation	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$)	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a < b$)
Eccentricity	$b^2 = a^2(1 - e^2)$	$a^2 = b^2(1 - e^2)$
Equation of major axis	$y = 0$	$x = 0$
Length of major axis	$2a$	$2b$
Equation of minor axis	$x = 0$	$y = 0$
length of minor axis	$2b$	$2a$
Vertices	$(\pm a, 0)$	$(0, \pm b)$
Foci	$(\pm ae, 0)$	$(0, \pm be)$
Equation of Directrices	$x = \pm a/e$	$y = \pm b/e$
Length of Latus -rectum	$2b^2/a$	$2a^2/b$

Ellipse →



Main facts about the Hyperbola

Equation	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $a > 0, b > 0$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ $a > 0, b > 0$
Eccentricity	$b^2 = a^2(e^2 - 1)$	$a^2 = b^2(e^2 - 1)$
Equation of transverse axis	$y = 0$	$x = 0$
Length of transverse axis	$2a$	$2b$
Equation of conjugate axis	$x = 0$	$y = 0$
Length of conjugate axis	$2b$	$2a$
Vertices	$(\pm a, 0)$	$(0, \pm b)$
Foci	$(\pm ae, 0)$	$(0, \pm be)$
Equation of Directrices	$x = \pm a/e$	$y = \pm b/e$
Length of latus-rectum	$2b^2/a$	$2a^2/b$



Circle

The equation of a circle with centre $C(a, b)$ and radius r is $(x - a)^2 + (y - b)^2 = r^2$

Equation of a circle: $x^2 + y^2 + 2gx + 2fy + c = 0$. Its centre $(-g, -f)$ and $r = \sqrt{g^2 + f^2 - c}$

***General equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents:**

- (i) Circle if $a = b$ and $h = 0$
- (ii) Parabola if $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$ and $h^2 - ab = 0$
- (iii) Ellipse if $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$ and $h^2 - ab < 0$
- (iv) Hyperbola if $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$ and $h^2 - ab > 0$

Introduction to Three-Dimensional Geometry

Distance, d , between (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Section Formula
$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$$

Midpoint Formula
$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

Coordinates of the **Centroid** of the triangle whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$.

Limits and Derivatives

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \quad ; \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad ; \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \quad ; \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad ; \quad \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a \quad ; \quad \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1 \quad ; \quad \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1$$

A function $f(x)$ is **continuous** at $x=a$ iff $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$.

A function $f(x)$ is **differentiable** at $x=a$ iff $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists finitely ie.

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h}.$$

Some Standard Derivatives

$$(i) \frac{d}{dx}(x^n) = nx^{n-1}$$

$$(ii) \frac{d}{dx}(\log_e x) = \frac{1}{x}$$

$$(iii) \frac{d}{dx}(e^x) = e^x$$

$$(iv) \frac{d}{dx}(a^x) = a^x \log_e a$$

$$(v) \frac{d}{dx}(\sin x) = \cos x$$

$$(vii) \frac{d}{dx}(\cos x) = -\sin x$$

$$(viii) \frac{d}{dx}(\tan x) = \sec^2 x$$

$$(ix) \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$(x) \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$(xi) \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$(xii) \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

$$(xii) \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

Product rule:

$$y = uv \text{ then } \frac{dy}{dx} = \frac{du}{dx}v + u \frac{dv}{dx}$$

Quotient rule:

$$\text{If } y = \frac{u}{v} \text{ then } \frac{dy}{dx} = \frac{\frac{du}{dx}v - u \frac{dv}{dx}}{v^2}$$

Probability

$$P(E) = \frac{\text{Number of favorable outcomes in } E}{\text{Number of possible outcomes in } S} = \frac{n(E)}{n(S)}$$

$$P(\bar{A}) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

Probability of occurrence of exactly one of A and B =

$$P(A) + P(B) - 2P(A \cap B) = P(A \cup B) - P(A \cap B)$$